

Federated Conjunction Assessment: A Framework for Reconciling Multiple Risk Assessment Solutions

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ABSTRACT

Proliferation of space situational awareness introduces a challenge: how should a spacecraft operator reconcile multiple, conflicting risk assessments for a single conjunction event? Lower-quality solutions with large uncertainties may inflate risk estimates and lead to unnecessary mitigation action while higher-quality solutions with lower risk estimates are discarded in the interest of conservatism. A decision framework is presented to systematically evaluate and reconcile multiple solutions for two satellites in a prospective conjunction. The framework operates in three phases: initial actionability assessment, statistical consistency among candidate solutions and final adjudication using direct-hit missed detection probability. The framework is applied to three historical case studies. The method provides a rigorous, scalable foundation for algorithmic decision-making in an increasingly federated space situational awareness paradigm.

1. INTRODUCTION

Burgeoning commercialization in the space situational awareness (SSA) industry presents the gift and curse of additional data to inform operational decisions. More data in principle should increase safety by ensuring a higher chance of identifying concerning close approaches. The conjunction assessment (CA) problem distills to a binary classification to either take or defer remediation action for a conjunction. When multiple solutions provide the same recommendation, the decision is simple; however, as is often the case, multiple solutions may contradict one another, leaving the operator in an ambiguous scenario. The collision probability (P_c) risk metric is driven by both the state estimate and the uncertainty for the primary and secondary objects involved in the conjunction. Counter-intuitively, the solution with larger uncertainty may be the one exceeding the action threshold, potentially compelling operators to act on lower-quality information. The burn size for sufficient mitigation may also be inflated when attempting to escape those larger uncertainties, as larger covariances typically require more aggressive avoidance strategies. As the number of orbit state estimates for the primary and secondary increases, the number of potential risk assessments increases as a square of the number of catalogs, which directly leads to an increased operational burden. An important contribution of this paper will be a technique to determine when divergent P_c estimates are an artifact of different covariance sizes versus two solutions that are inconsistent with each other, in which case at least one solution must be incorrect.

The driving forces behind different risk solutions are divergent orbit states and covariances for one (or both) of the objects involved in a conjunction. At a minimum an owner/operator (O/O) of a satellite may need to decide between their own solution and the non-cooperative, ground-tracking solution provided by the Space Surveillance Network (SSN). Recent developments in the commercial sector, including the Office of Space Commerce's (OSC) pathfinders to explore a fully commercial catalog and SpaceX's recently announced Stargaze capability will only increase this complicated decision process. In the absence of a fused orbit determination result, an operator needs to decide how to act on this data.

There are several ways to approach this problem. One solution is to simply maneuver against all data sources; operators would probably default to this approach, and large constellations with replaceable individual satellites seem to prefer

it, but it causes increased cost on more traditional operations. Another option is to identify when two states are consistent and fuse them, as presented in [1]. Another technique is to search for a consensus among multiple solutions relying on the decentralized networks to converge in the aggregate [2]. A paper by Hu et al. [3] gives a good overview of the variety of consensus and fusion techniques available, which are largely derived from techniques in the broader tracking field. Fundamentally, the key question is: how can I reduce the number of P_c estimates upon which I need to act? Consensus and fusion are options in achieving that end. A core goal of this work is to create an algorithmic solution to simplify and, where possible, eliminate confusion from multiple orbit solutions in the lead up to a conjunction event. This work presents an extensible framework for systematically evaluating and reconciling multiple orbit determination (OD) solutions for the same conjunction event. The proposed framework aims to provide operators with a rigorous methodology for adjudicating between conflicting CA recommendations through a multi-stage assessment process which uses statistical consistency testing and OD quality assessment. This framework is not incompatible with the other techniques presented but instead grounds them in something that an operational program can implement in the day-to-day.

Determining upon which P_c to act is better framed as choosing which state and uncertainty estimate is most reliable for the primary and secondary satellites. The perhaps simplest way to reconcile multiple solutions is to eliminate OD results with poor characteristics; if two solutions disagree but one has clearly superior OD quality, the reconciliation can be simplified. The process begins by isolating the primary and secondary objects' OD solutions from all available conjunction data messages (CDM). An actionability assessment is performed on each unique primary and secondary object OD solution to determine if the data are of high enough quality to use for CA. The NASA Conjunction Assessment Risk Analysis (CARA) program developed actionability criteria which include checks for covariance quality and OD fit quality among other data quality indicators. While absence of ancillary OD information at this stage is not disqualifying, any solution with documented actionability violations will be discarded. Heuristic composite OD quality scores developed at the NASA CARA program are attached to each solution. These quality scores will be used in later stages to adjudicate between statistically inconsistent solutions.

An important further refinement presented in this paper is a technique to determine when OD solutions are consistent with each other. We compute the 3 degree-of-freedom (DOF) position Mahalanobis distance between each unique solution relative to the combined covariance. Given both solutions involved in a Mahalanobis distance calculation provide realistic covariances, the distribution of squared Mahalanobis distance should follow the χ^2 distribution with 3-DOF. A hypothesis test is constructed with the null hypothesis that the solutions are statistically consistent (i.e., estimate the same truth state with statistical accuracy). The known χ^2 distribution is used to produce a P-value indicating the probability of observing the computed Mahalanobis distance (or one larger) under the null hypothesis. A P-value threshold is established (e.g., $P > 0.02$ for 98% confidence). If any solution pair fails the hypothesis test, indicating statistical inconsistency, the solution with the lowest OD quality score is discarded and the test is re-run with the remaining solutions.

For solutions that pass statistical consistency testing, covariance sizes are compared. If solutions are statistically consistent but covariance sizes differ significantly, the solutions are ranked by their missed detection probability (P_{md}) [4] with the lowest, most confident solution preferred, as it provides a more precise risk estimate without sacrificing statistical validity. If P_{md} performance is comparable and solutions remain consistent, either solution may be used, or a weighted combination can be employed.

The expected outcomes include:

1. An algorithmic approach that accepts multiple CDMs and automatically computes statistical consistency metrics and quality scores.
2. A validated decision tree for solution selection applicable to operational scenarios.
3. Case studies that demonstrate the framework applied to real historical CARA conjunction events.

The methodology is directly applicable to any multi-source CA environment and is scalable to scenarios involving more than two solutions making it highly relevant to the evolving federated SSA ecosystem where multiple commercial and government entities contribute to space traffic coordination.

2. CHALLENGES TO FEDERATED OPERATIONAL RISK ASSESSMENT

To reiterate the core challenge being addressed in this paper, what does an operator do when multiple orbit determination solutions (typically presented as ephemerides) give different risk assessment results? This is not hypothetical; NASA CARA experiences many such events routinely where a decision must be made between an ephemeris produced with the best-in-world modelling at the 18th Space Defense Squadron (SDS) or the more mission-specific knowledge produced by a NASA O/O. We also experienced an event with heavy disagreement between our internal data and an ephemeris furnished by SpaceX. This paper will not comment on the relative quality between SpaceX and CARA data but instead simply state that, at a minimum, we need the ability to react to data from a system such as Stargaze.

Federated CA, as a topic, suffers from a lack of clear definitions, so this paragraph attempts to lay out some explicit terminology. **CDM/ P_c selection** is the process of choosing upon which CDM and corresponding P_c an operator should react. This is the end to which an operator needs to arrive but does not best describe the technical problem. Instead, **Ephemeris Selection** is the technical step required to choose a P_c . We choose the most trustworthy ephemerides which give a resulting CDM/ P_c . **Ephemeris Actionability** is best defined by their opposite. Unactionable data is data upon which we should not act due to clear technical problems. **Synthetic CDM** is a CDM that is synthesized from data from two existing CDMs. This is relevant when the selected primary/secondary pair do not exist in a single CDM and is discussed in **Section 4.2**.

Furthermore, federated CA as a topic has a wealth of inherent technical challenges that move beyond the relatively simple selection of *in situ* maneuvers. New data sources must be vetted for reliability and robustness in an operational context. Unvetted data sources could degrade operational safety, either maliciously or by mistake. A separate question asks when multiple solutions should be merged, either at the observation level or through state fusion. Fusion should lead to a better solution, but care needs to be taken to not corrupt good data. Several topics the world faces primarily concern software development: as the number of data sources increases costs, complexity, and security practices need to keep pace. Finally, operations already experience the difficulty of explaining the nuanced statistics of conjunction risk assessment under stressful circumstances. As the required techniques increase in complexity, the operators, education and communication need to keep apace.

The variety of challenges often confuse conversations around federated CA as it becomes unclear which problems are and are not being solved. To frame the scope of this challenge, we present a consolidated list of potential operational risks that necessitate future research and mitigation:

Table 1: Enumerated components of the federated CA problem.

1 Ephemeris Selection	This is the primary topic being addressed in this paper and is discussed thoroughly throughout.
2 Data Sourcing	The process of going through, selecting and understanding the data types that will be considered viable and acquired for operations.
3 Covariance Realism	This may be the best way to source actionability but requires large amounts of data and coordination with those being evaluated.
4 Data Fusion	Multiple orbits for the same object can be merged into a single best estimate.
5 Big Data	When fusion is unavailable, the number of ephemeris data sources expands the number of P_c options exponentially.
6 Operator information overload	The production of so many P_c estimates and their sources make operations muddled, laborious, and confused.
7 Explanation to Decision Makers	The technical challenges can lead to unintuitive solutions which need to be understood and be explained to satellite stakeholders.
8 Event Coordination	On top of who maneuvers, coordination of the criteria for event mitigation needs to be agreed upon.
9 Inter-Center Agreement	Safety centers will need to develop new strategies for how to coordinate questions of safe operations.

10 Cyber Security	The enterprise is based on rapid decision and flow of trusted information, which can be exploited.
11 Novelty of Problem	Federated CA has arrived before any operational solution could be tested, which may harbor unknown risks.

The purpose of this section and Table 1 is to demonstrate the scope of the problem and enable the scope of this paper to be better defined. This paper is primarily focused on topics 1-3. Of the other topics, data fusion has been discussed in literature and is a parallel approach to the results presented here. Several of the remaining topics are being worked parallel to this effort on CARA. However, some of the remaining topics require more attention from Space Situation Awareness (SSA) policymakers but ultimately can be informed by the technical content of this paper.

3. EPHEMERIS ACTIONABILITY ASSESSMENT

The ephemeris actionability assessment is applied to each unique solution to understand the baseline validity of the solution. The actionability assessment acts as a filter as to whether a solution should even be considered for risk assessment. The ethos of the actionability assessment process is to exclude solutions only when there is clear evidence of poor quality. If the assessment finds a solution to be ambiguously actionable or perhaps lacks the necessary metadata to decide, the solution is allowed through this gate. In the current operational paradigm, this is not a problematic stance, although as the number of ephemeris providers increases, NASA CARA may need to re-evaluate this assumption and provide a barrier to entry for OD metadata availability and baseline performance.

3.1 Definitions of Actionability of an Ephemeris

NASA CARA developed actionability criteria primarily with the intent of evaluating secondary orbit determination solutions for actionability [5]. In CARA's experience, there will always be secondary objects, derived from the Astrodynamics Support Workstation (ASW) or commercial systems, with tenuous OD [6]. The problematic secondary solutions originate from sparsely tracked debris with unfavorable viewing geometry for the SSN. The same actionability filters are applied to primary OD solutions although they rarely issue violations of the established criteria. CDMs generated using ASW solutions and provided to NASA CARA contain a wealth of metadata relating to the OD fit for each object. The exact algorithm in use is publicly available at the NASA CARA GitHub site.¹

3.2 Covariance Realism

CARA processes ASW solutions for primary and secondary objects. Additionally, most NASA CARA assets furnish O/O ephemerides derived from coherent tracking or global navigation satellite system (GNSS) based solutions. With increasing prevalence, secondary payloads also furnish O/O ephemerides either directly to the Space Force or through intermediaries such as the Office of Space Commerce or SpaceX screening service.

Of tremendous importance to the CA enterprise is the realism of the state and uncertainty represented by the covariance matrix delivered along with these O/O ephemerides. In the context of CA, uncertainty or covariance realism is the expectation that the predicted covariance matrix accurately describes the distribution of state errors produced by the state predictions. Exact evaluation of covariance realism has taken a few forms in literature but generally involves evaluation of the Mahalanobis distance between predicted state and covariance and an *ex post facto* definitive reference, often a precision orbit determination solution. Some detailed studies have shown different methods of evaluating the assumptions [7] [8] [9] [10].

The ASW solution and its associated covariance have received attention in terms of quality assurance and generally perform well [11]. The quality is also reasonably constant across the catalog with some variation due to viewing geometry, susceptibility to drag, and unknown secondary maneuver capabilities. In contrast, the variety of ephemeris quality provided by NASA and external ephemeris providers is wide. The status quo is to accept any ephemeris at face value and assume diligent covariance realism analysis and quality assurance is on-going. In practice, CARA observes some O/O ephemeris providers tend to devote more resources to covariance realism than others. Lack of

¹ https://github.com/nasa/CARA_Analysis_Tools/tree/master/DistributedMatlab/ODQualityAssessment

covariance realism in some O/O predictions led CARA to include metrics on covariance realism into the ephemeris selection process.

The prominent philosophy on covariance realism is that past performance predicts future behavior. This philosophy manifests in covariance realism with an evaluation scheme that operates on a moving window of past performance. Sufficient for most spacecraft operations is a look-back period of the last 90 days of predicted ephemeris generation corresponding to roughly three solar rotation periods. Rigorous evaluation methods require some form of “ground-truth” against which predictions will be evaluated. In this work, we make no assumption a ground-truth exists for a given object. We instead evaluate covariance realism in the form of an overlap analysis. Rather than asking how well predictions represent the state error distribution around a definitive reference, we ask how well each prediction represents the distribution of the last prediction for a given event. Conveniently, the overlap analysis can be done solely with states and covariance matrices extracted from subsequent CDMs. Fig 1 demonstrates schematically the comparison technique used.

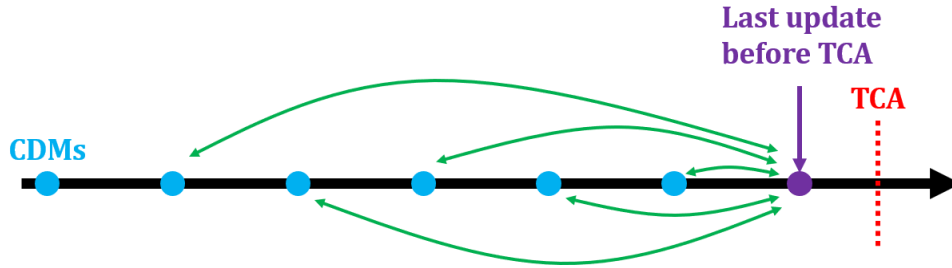


Fig 1: CDM overlap comparison schematic for covariance realism assessment.

Small two-body propagations adjust all CDMs to a common time of closest approach (TCA). From there, position Mahalanobis distances for the primary and secondary objects can be computed as follows:

$$MD^2 = (x_{LU} - x_{PD})^T (C_{LU} + C_{PD})^{-1} (x_{LU} - x_{PD})$$

Where x_{LU} is the position vector extracted from the last-update CDM before TCA and x_{PD} are all position vectors from previous, predicted CDMs leading up to TCA. Likewise, the covariance matrices C_{LU} and C_{PD} are the last-update and predicted position covariance matrices.

The ultimate output of the covariance realism assessment is a Mahalanobis distance cumulative distribution function (CDF) binned by propagation time. Example CDFs for a sample mission are plotted in Fig 2.

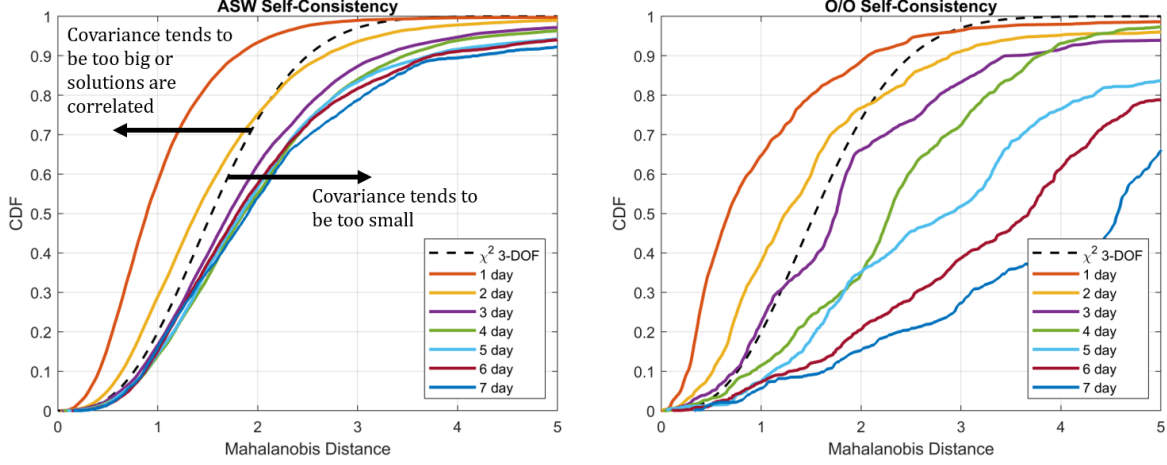


Fig 2: Mahalanobis distance cumulative distribution functions for a sample mission.

Using this approach, solutions are assumed to be correlated in that subsequent ephemeris predictions are not statistically independent. Correlation manifests as the CDF skewing left of the χ^2 3-DOF CDF. It is especially evident for short propagations when the OD completed for x_{LU} and x_{PD} likely shares several observations. Currently, there is no penalty for heavily correlated solutions. A quality score is derived from the point at which the empirical CDF crosses the hypothetical χ^2 CDF. The two ephemeris providers in Fig 2 at a 1-day propagation length score 99% and 96% for ASW and O/O respectively. The scoring approach is rudimentary but is intended to give an idea of how consistent this ephemeris provider will be in predicting its last update. Given no accounting is done for correlation, this should be a low bar to cross, especially for shorter propagations. More rigorous evaluation approaches are in development and use at CARA, but a reliable source of truth must be identified for all objects to leverage those methods.

Implicitly assumed in the overlap approach is that the subsequent predictions will approach the state of the ground-truth. It does not capture systematic bias which may alter even the final prediction substantially. To capture the potential for bias, we can assume that an ASW solution generated within 12 hours of the TCA will sufficiently approximate the truth. Thus, two covariance realism evaluations can be performed for non-ASW solutions as needed. Generally, these are used for reference and do not directly enter the decision-making process. This evaluation method is used on a case-by-case basis when the ASW quality can be accurately assessed.

4. EPHEMERIS SELECTION PROCESS

Assuming multiple ephemerides for the primary and/or secondary object passes the actionability filter, the process must now winnow down actionable ephemerides.

4.1 Ephemeris Consistency

Ephemeris consistency is evaluated in real-time in the presence of disparate P_c estimates. Of most interest to decision-makers are those situations in which two or more P_c estimates straddle an action threshold. This situation is not necessarily problematic. Imagine the scenario of a sparsely tracked primary object in an equatorial orbit. The SSN may have limited viewing opportunities and therefore produce a solution with large uncertainties at epoch. Conversely, an O/O may use a high accuracy, GNSS solution and provide an ephemeris with much smaller uncertainties. It is easy to imagine the SSN-derived solution resides in the so-called “dilution” region wherein the P_c estimate is diminished by large uncertainties [12]. Conversely, the O/O solution may exceed the action threshold due to relatively low uncertainty. Ephemeris consistency evaluates the Mahalanobis distance between the two solutions to determine if they are statistically consistent, albeit with divergent P_c estimates. The squared 3-DOF position Mahalanobis distance (MD^2) between the two distinct primary object position vectors (x_1 and x_2) relative to the combined position covariances (C_1 and C_2) is computed as follows:

$$MD^2 = (x_2 - x_1)^T (C_1 + C_2)^{-1} (x_2 - x_1)$$

Resultant squared Mahalanobis distances are compared against the χ^2 CDF with 3-DOF to determine a P-value. The P-value represents the probability of observing the given squared Mahalanobis distance or one larger under the null hypothesis that both solutions are estimating the same truth state. CARA compares the P-value against a threshold of 0.02 indicating a 2% chance of observing a given MD^2 value or greater by random chance alone.

If the two or more solutions are statistically consistent, it is easier to state that no P_c estimate is “wrong” per se, and rather they span a reasonable range of P_c estimates for this given conjunction.

4.2 Synthetic CDM Creation

Synthetic CDM creation is required when the “best” solutions for the primary and secondary objects come from different screening services and therefore no common CDM exists from which to compute a P_c . Imagine two screening services. Both screening services key off their independent catalogs, along with any submitted Owner/Operator (O/O) data. For an event, the primary submits to both screening services while the secondary only submits to the first service. Such situations create a network of potential but non-exhaustive CDMs as seen in Table 2. If, in this hypothetical scenario, the secondary was performing a maneuver only captured in their O/O solution while the primary is uniquely well tracked in catalog 2, no system would produce the “best” CDM. In current CARA operations, this is an exceedingly rare scenario. If it were to occur, CARA would request the best ephemerides both be submitted manually for screening in the service of our choice. However, as commercial, for purchase catalog data becomes more commonplace we expect situations where such a request is infeasible or simply logistically difficult. CA providers should consider what requirements are necessary to extract states from different CDMs and propagate them to a common TCA. In most conjunctions, two-body or even rectilinear assumptions would likely be valid to cover the new TCA. Those conjunctions with low relative velocity might require further study to develop an appropriate approach.

Table 2: Potential permutations of CDMs and primary/secondary solutions in a federated CA landscape.

	Primary (O/O)	Primary (Catalog 1)	Primary (Catalog 2)
Secondary (O/O)	CDM _{O,O}	CDM _{1,O}	CDM _{2,O}
Secondary (Catalog 1)	CDM _{O,1}	CDM _{1,1}	No Data
Secondary (Catalog 2)	No Data	No Data	CDM _{2,2}

Another attractive synthetic approach is to fuse solutions from different ephemeris providers to, in theory, produce a composite solution superior to either constituent solution. In fact, solution fusion has been discussed and attempted in literature [1]. Data fusion remains a compelling area for future research, but the ephemeris selection procedure is favored in the current operational paradigm for its immediate, computationally simple, and transparent implementation.

4.3 Other Considerations for Ephemeris Selection

A common scenario in CARA’s operational experience is one in which multiple competing solutions for the primary object pass all actionability assessments and contain statistically consistent state estimates. The user is left with two P_c estimates and must make one decision: take mitigative action or do nothing. This work offers the missed detection probability (P_{md}) devised by Sweetser et al. as the adjudicating metric for these scenarios [4]. The P_{md} is computed with the small-hard body radius (HBR) or large-covariance approximation as follows:

$$P_{md} = 1 - \max\left(1 - 2 \frac{T_a s_x s_y}{R^2}, 0\right)$$

Where s_x and s_y are the semi-major and minor axes of the combined covariance as projected on the conjunction plane, R is the combined HBR, and T_a is the action threshold, which is 10^{-4} for most CARA-supported missions.

Both solutions are evaluated for their direct-hit missed detection probability and the solution with a lower missed detection probability is favored. In effect, this adjudication favors the smaller covariance as projected on the conjunction plane. The highest- P_{md} solution can be thought of as the one with the most confidence in decision making. It is the natural selection but should not be done without considering the covariance realism of the ephemeris provider as described in **Section 3.2**. Using P_{md} alone as a selection criterion could favor those ephemeris providers with “smug” navigation filters whose covariance is overly confident and produce an under-sized representation of the actual error distribution. More often than not, under-sized covariances produce under-estimates of the P_c which in turn results in under-reaction to risky conjunctions.

An alternative approach would be to act on the higher of the P_c estimates. This is viable but neglects the benefit of a better-quality solution. In the case where both solutions are in the so-called “robust” region, the lower-quality solution is likely to produce the higher P_c estimate [12]. In the interest of conservatism, we would not dissuade an operator from acting on the higher of the two P_c estimates, especially if the missed detection rates are very similar.

4.4 Ephemeris Selection Flowchart

Fig 3 illustrates the complete algorithm for ephemeris selection, which fuses the concepts discussed previously into one comprehensive pipeline. The framework operates in three distinct phases separated by colored boxes with labels for each phase underneath. The entire flow chart is applied to both the primary and secondary solutions from all candidate CDMs for a given conjunction.

Phase I is the initial actionability filter which is applied to every unique object solution. Before any cross-comparison takes place, each solution must pass the basic actionability criteria check. Solutions with critical flaws such as non-positive definite covariance matrices, severe tracking degradation or unusably large covariance matrices are discarded at this stage. It is worth noting any solution that lacks OD metadata such as those provided by external O/O is allowed through this gate. An ephemeris provider may perceive this as a perverse incentive to exclude OD metadata to bypass the actionability check and reach farther into the flow chart. It may warrant investigation as to whether certain OD quality parameters should be standard metadata included with the delivery of every ephemeris.

Phase II is split into two sub-phases although all solutions begin with Phase II A: an internal Mahalanobis distance calculation between all actionable, candidate solutions. Statistical consistency is deemed to be any P-value exceeding the 0.02 threshold. If all solutions within a cohort exceed the P-value threshold, they are deemed statistically consistent and exit the comparison block to the right, proceeding to Phase III. If any solution pair is deemed statistically inconsistent, the flow proceeds down to Phase II B. In this section the solutions are ranked by their OD quality score, and the lowest performing solution is discarded. With one fewer solution, statistical consistency is again checked and that loop is repeated until either statistical consistency is achieved or only one solution remains.

Finally, Phase III A is reached when multiple, statistically consistent solutions are identified. We are now in a scenario where we have multiple reasonable estimations of the spacecraft state and uncertainty and therefore multiple reasonable estimates of the collision risk in the form of multiple P_c estimates. The Sweetser P_{md} is used as the final adjudicator as the test searches for the solution with the lowest P_{md} . The lowest P_{md} represents the solutions with the smallest uncertainty as projected on the conjunction plane which ideally correlates to the solution with the best quality information. Phase III A is designed to avoid the dilemma of the poorer-quality solution recommending action while a better, yet statistically consistent solution, indicates no action is required. If the remaining solutions straddle an action threshold, an operator can reasonably react to the solution that suggests action to err on the side of caution rather than further winnow down solutions. CARA pursues the method of choosing the lowest P_{md} because often the ASW solution contains more recent observations and orbit determination than competing solutions. The newer solution with a lower P_{md} can therefore portend a fall in P_c for all solutions, it simply has the advantage of lower latency.

Phase III B is the less satisfying conclusion of the process wherein solutions are discarded until only one remains, which is selected as the best solution simply by its perseverance.

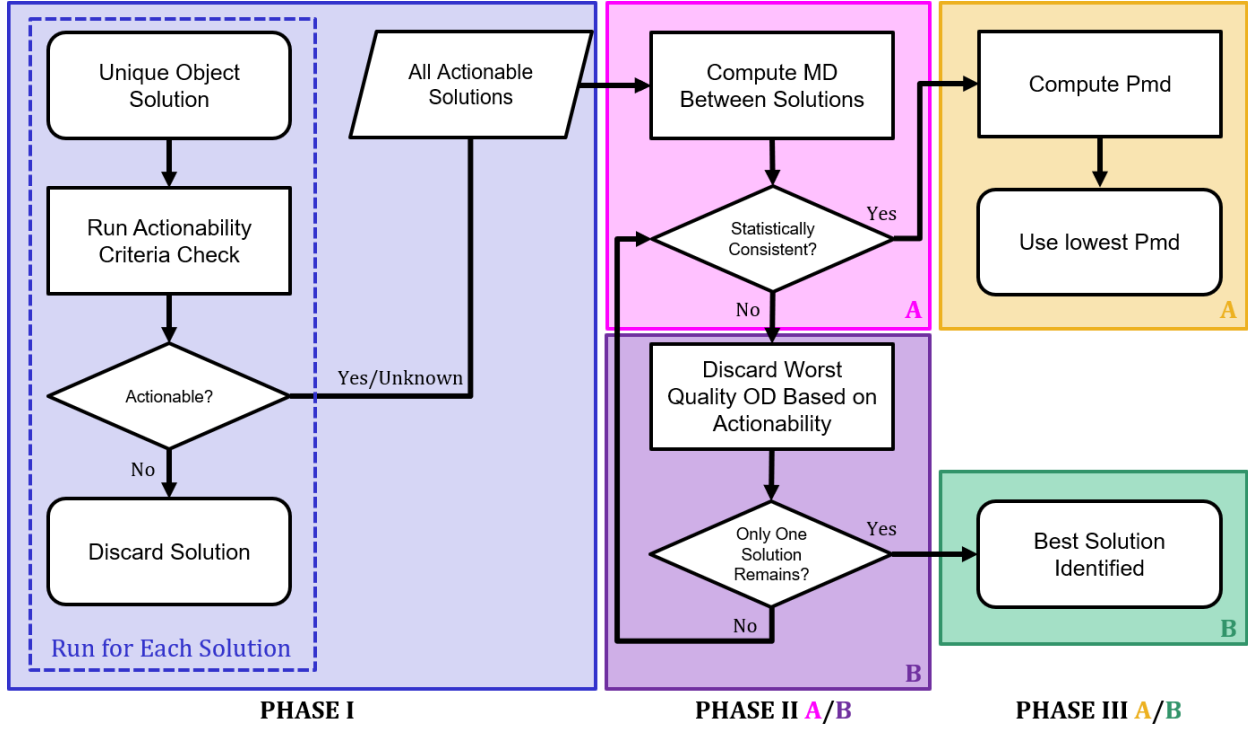


Fig 3: Ephemeris selection flow chart.

The goal of the framework presented here is to arrive at one, single object pair for use in risk assessment. However, a practical caveat is warranted: winnowing down the pool of potential solution pairs strictly to one may not always be necessary. For example, two statistically consistent solutions with comparable missed detection probabilities may still straddle an action threshold. Similarly, it may be ambiguous which of two inconsistent solutions has less reliable OD. It may be prudent to simply accept the conservative estimate in this scenario. While the framework provides a mathematical justification to break the tie, both solutions represent a valid understanding of the physical state. In this scenario, rather than relying on a minor difference in P_{md} to waive a maneuver, it may be prudent for operators to simply exercise conservatism, accept the higher P_c estimate, and proceed with mitigation.

5. RESULTS AND CASE STUDIES

The flow chart developed in **Section 4** is now applied to a series of case studies drawn from recent CARA operations. The cases studies will demonstrate the utility of the ephemeris reconciliation process on real conjunctions with divergent P_c estimates. The figures follow a three-panel layout. First is the P_c update sequence for the entire event with both O/O primary vs. ASW secondary and ASW primary vs. ASW secondary conjunction results. The second panel contains the position Mahalanobis distance between the primary solutions produced by the ASW and O/O in the same screening. The first two panels are available to operators in real-time when analyzing events. The third panel is an after-the-fact product comparing each predicted primary solution with a definitive OD solution produced after TCA has passed. The third panel along with any scatter points after the blue rectangle represents the *postmortem* analysis which is not available to operators in real time. The blue rectangle highlights the screening result being considered for ephemeris reconciliation.

5.1 Case Study I

Case Study I is presented in Fig 4. The update of interest is roughly 2 days prior to TCA when the mission is beginning to plan a risk mitigation maneuver (RMM) for the event. As evidenced in the top panel, the O/O vs. ASW solution yields a P_c that exceeds the maneuver threshold of 10^{-4} . The last five updates produced over the previous 24 hours also show a red-level P_c in the O/O vs. ASW solution, but only a moderate risk in the ASW vs. ASW solution. The second panel of Fig 4 shows the Mahalanobis distance between O/O and ASW primary solution pairs in each

screening. Notably, all but two updates thus far in the update sequence indicate statistical inconsistency (i.e., P-value below 0.02) between the ASW and O/O primary solution. Both solutions are deemed actionable with the O/O solution receiving a lower, yet acceptable score for covariance realism.

Reviewing the framework in Fig 3, both primary object solutions pass the basic actionability checks in Phase I. In Phase II A, statistical inconsistency is noted as seen in the middle panel of Fig 4. The loop in Phase II B is engaged to discard the solution with a lower actionability score. In this case, the O/O solution scores lower in the covariance realism metric and is therefore discarded. With only the ASW primary solution remaining, the ultimate recommendation is to defer to the ASW vs. ASW P_c estimate which does not exceed the action threshold of 10^{-4} .

Considering updates that were received after the analyzed update, both P_c ultimately reduce to effectively zero within one day of TCA. The third panel of Fig 4 shows the comparison of each predicted solution with the definitive OD solution for the primary. The poor O/O covariance realism performance is reflected here in the form of large Mahalanobis distances, well above the the 0.02 P-value threshold established at CARA. In this case study, operators were allowed to dismiss a red P_c estimate based on the historical covariance realism performance of the O/O. In retrospect, this decision was correct as the O/O predicted solution never approached the definitive solution. Effectively, the proposed framework avoided an over-reaction to a relatively unreliable P_c estimate.

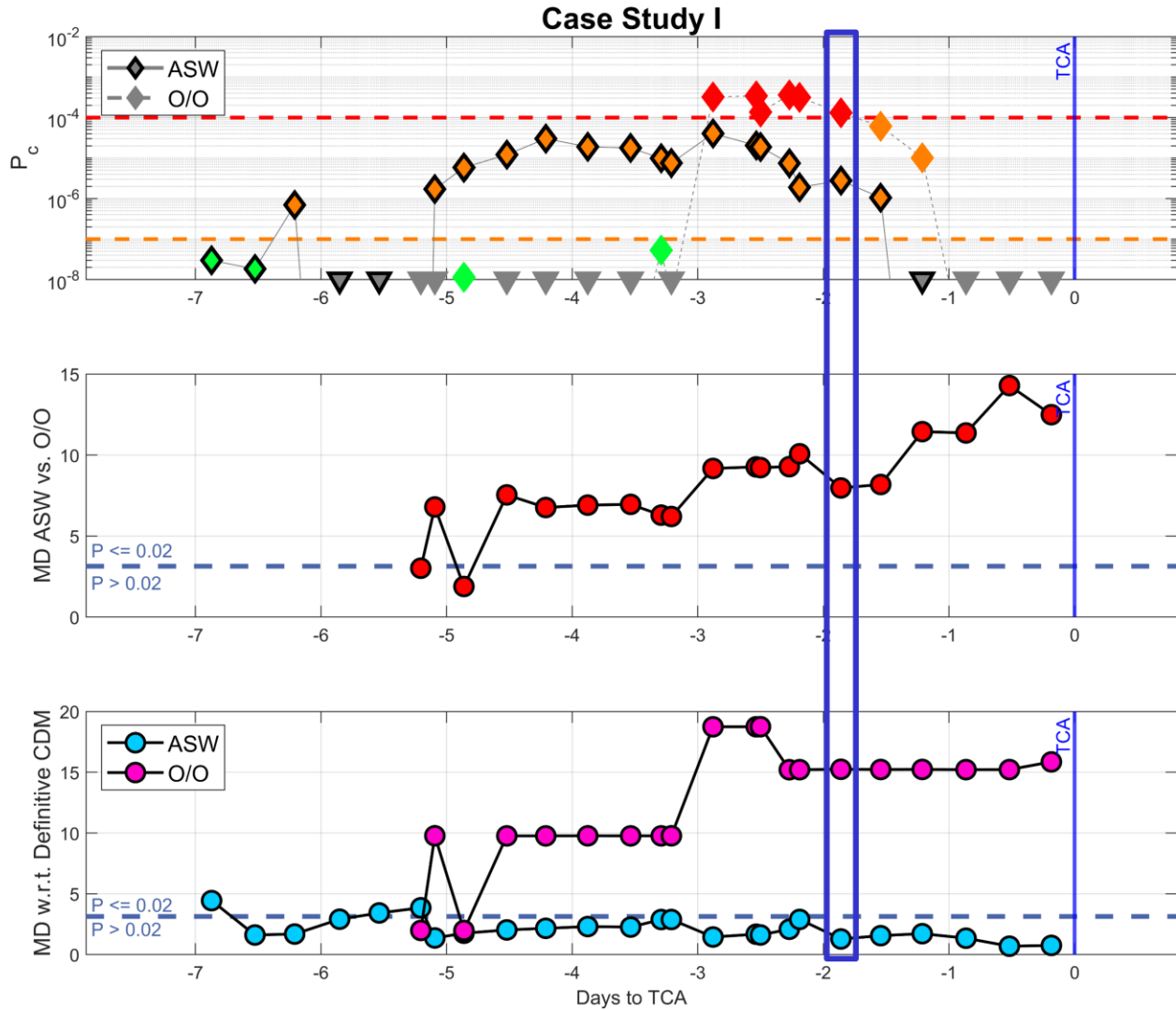


Fig 4: Case Study I update sequence and statistical consistency measures.

5.2 Case Study II

Case Study II is presented in Fig 5. In this conjunction, the update of interest is less than 1 day to TCA and mission operations teams wish to understand if a maneuver is required based on a relatively late notice P_c estimated in the ASW vs. ASW solution. Statistical consistency is achieved throughout the update sequence between the O/O and ASW primary solutions as evidenced by the middle panel.

Tracing this case study through the flow chart in Fig 3, both solutions are deemed actionable in Phase I, statistical consistency is proven in Phase II A, so now Phase III A is engaged to compute the missed detection probability of both solutions. The O/O vs. ASW produced a missed detection probability of 0.9% compared to the ASW vs. ASW missed detection probability of 17.7%. Following the flow chart, the O/O vs. ASW solution is therefore deemed the best solution as it has more decision authority with its lower missed detection probability. The mission is therefore advised to stand-down maneuver planning with the expectation that the ASW vs. ASW P_c will imminently fall as new observations and shorter propagation times reduce the uncertainty.

Reviewing the single update after the blue rectangle and the third panel, the decision process appears to have paid off. In fact, the ASW vs. ASW P_c estimate did fall below the action threshold in the next update. The missed detection probability of both solutions fell once more with the O/O vs. ASW solution ending at 0.2% and the ASW vs. ASW falling to 6.4%. The third panel shows that within 3 days of TCA, both ephemeris providers delivered solutions that were statistically consistent with the definitive OD solution. The ephemeris reconciliation process allowed the operator to avoid superfluous maneuver planning. Indeed, this scenario is one in which the quality of the SSN-based tracking on the primary object is diminished by inconvenient orbit geometry. The O/O solution provided a more confident solution for risk assessment and the ephemeris reconciliation process worked as designed.

The difference in missed detection probability is best demonstrated by inspecting the combined covariances of the various solutions as projected on the conjunction plane. Fig 6 shows the two conflicting solutions analyzed in magenta and light blue along with the final O/O update in blue. The conjunction plane here is defined relative to the ASW solution with the relative velocity out of plane and the miss vector placed on the horizontal axis. From the conjunction plane vantage point, it is clear the O/O provides much higher certainty while not presenting solutions that are statistically inconsistent with the ASW solution. Combining statistical consistency testing and missed detection probability provides a mathematical justification for choosing the O/O solution which appears to be the intuitive choice when inspecting Fig 6.

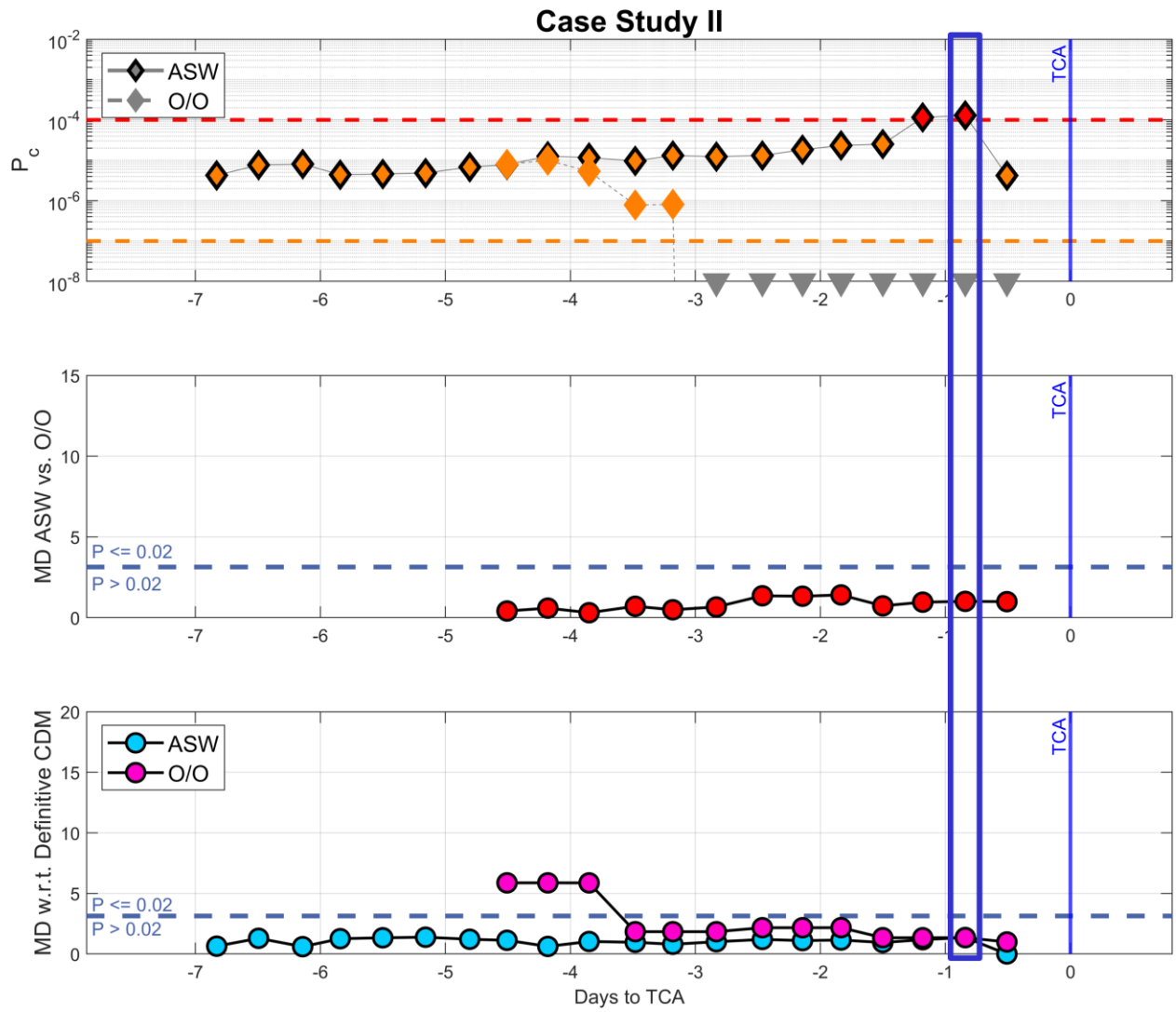


Fig 5: Case Study II update sequence and statistical consistency measures.

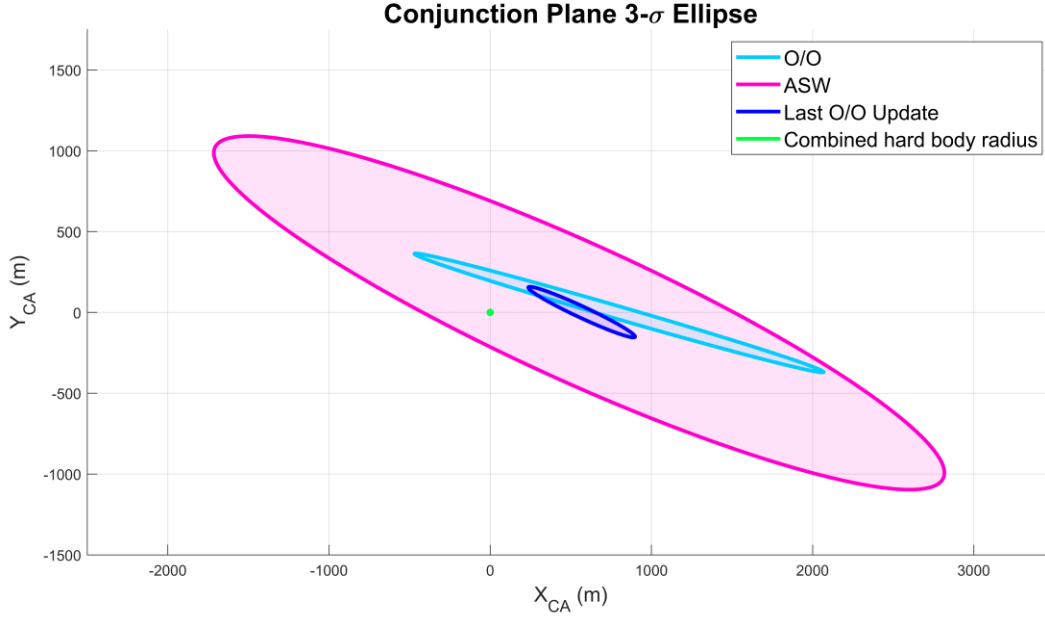


Fig 6: Conjunction plane diagram of the various solutions of interest in Case Study II.

5.3 Case Study III

Case study III is presented in Fig 7. The update of interest is just within 2 days of TCA and operators are tracking a persistent and actionable red level P_c in both the ASW vs. ASW and O/O vs. ASW solutions. In the latest update the O/O vs. ASW solution fell below the maneuver threshold, but the ASW vs. ASW P_c remains red. Statistical consistency is achieved throughout most of the update sequence, but the two primary object solutions diverged in the 24 hours prior to the update of interest.

Reviewing the flow chart in Fig 3, both primary object solutions are deemed actionable and exit Phase I with the O/O primary object solution receiving a lower covariance realism score than the ASW solution. In Phase II A, statistical consistency is not achieved, leading to disposal of the O/O solution. With only the ASW primary solution remaining, the resultant ASW vs. ASW solution is considered the primary risk assessment solution which indicates maneuver planning should continue.

As seen in the *postmortem* updates, the ASW vs. ASW P_c estimate did indeed fall in the subsequent updates. Both P_c estimates arrived at effectively zero within one day of TCA. On the surface, it might appear that the ephemeris reconciliation framework needlessly delayed waving the maneuver and mandated operators continue unnecessary planning. However, the third panel in Fig 7 demonstrates the ephemeris reconciliation framework's value: it prevented operators from making the right decision for the wrong reasons.

While the O/O vs. ASW P_c estimate dropped below the threshold sooner, the bottom panel demonstrates that the O/O primary object solutions were not statistically consistent with the definitive solution. The O/O solution provided a convenient P_c in that it dropped below the action threshold, but it was fundamentally inaccurate. The ASW primary solution, on the other hand, shows excellent statistical consistency with the definitive truth throughout the entire update sequence. By strictly adhering to the flowchart, the algorithm successfully identified and discarded degraded data, preventing operators from cherry-picking a lower P_c . This case study demonstrates that the framework inherently errs on the side of statistical rigor, ensuring that mitigation efforts are only stood down when physically accurate data supports doing so.

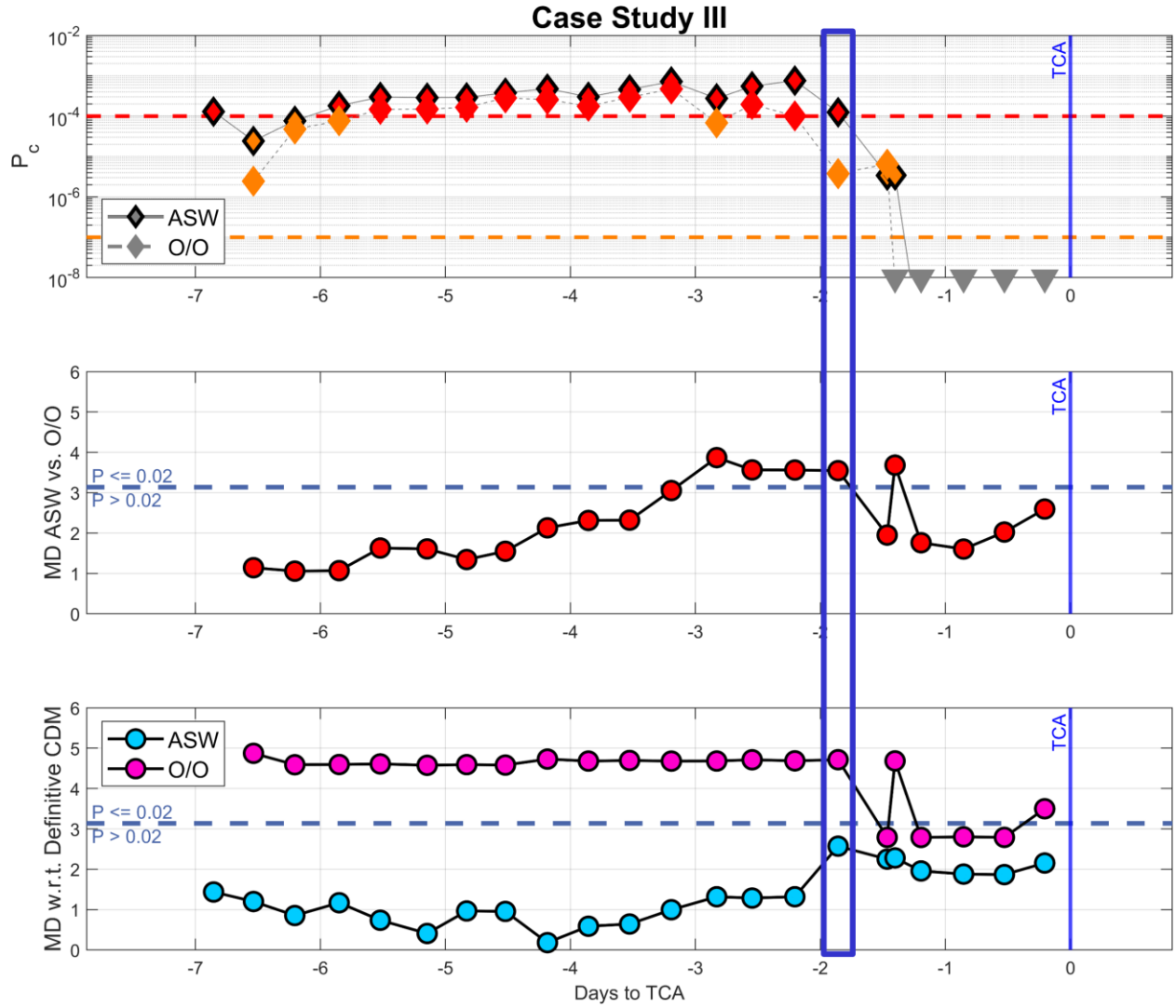


Fig 7: Case Study III update sequence and statistical consistency measures.

6. CONCLUSIONS

NASA CARA strives to take prudent measures at reasonable cost to improve mission space flight safety. Sometimes this means actions that are onerous to normal satellite operations while sometimes this means minimizing the operational burden on satellite operations. The results presented in this paper attempt to straddle this line. In some cases, the proposed methodology will recommend that no mitigation maneuver be planned in the face of a higher P_c because we believe there is better data. In some cases, it may be forced back to a more conservative default. It should be noted that as of publication the presented decision flow is only used at CARA with heavy human presence within the decision process. It is not a current automated algorithm. We face, like many others, an increasing rate of events and data sources, will likely need to implement more automation, and so these results represent a blueprint for future automation. This will be done with testing that will likely shape some elements of the presented process. As we embark on this effort, the goal of this paper is to stimulate conversation and feedback on the best techniques available. Federated CA, in its full form, is much more complicated than the algorithms presented here, but a consensus among operators on acceptable decision processes will make future policy work much easier.

We propose a framework for solution reconciliation in the event of multiple risk assessment solutions for spacecraft collision avoidance. The framework leverages NASA CARA experiential knowledge on OD quality, predicted

trajectory, and covariance realism while integrating statistical consistency and missed detection probability as further adjudicators.

Several historical conjunctions are presented as case studies that exercise the various paths of the framework. With the addition of after-the-fact OD updates, we find the framework successfully directs operators to the best risk assessment solution for a given conjunction. Crucially, the retrospective analysis demonstrates that the framework prevents unnecessary maneuvers driven by over-sized covariance matrices while also guarding against the temptation to defer action based on artificially low P_c estimates derived from unrealistic, under-sized covariances.

While ephemeris selection is the current operational preference, further research into state and covariance fusion remains a compelling avenue for generating synthetic CDMs, provided cross-correlation challenges can be adequately modeled. Ultimately, this framework provides a necessary foundation for safe, scalable decision-making in an increasingly federated SSA ecosystem.

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